Module 4 slides: Clearance and Elimination (worked examples)

Module 4 - Drug Elimination and Clearance · Lecture slides, worked examples · 14 Sep 2026

Name _____ Date _____

Five hundred mg of a drug was administered by rapid IV injection. The V_D is 15 L and the elimination half-life is 8 hours. Urine samples were collected for 48 hours and 300 mg of unchanged drug was recovered.

Slide 22 What fraction of the dose is excreted unchanged in the urine?

Slide 22 Calculate k.

Slides 10 and 22

Calculate k_e.

Slide 22 Calculate Cl_T.

Slides 9 and 22

Calculate Cl_R.

Slides 11 and 22

Calculate Cl_H.

Slides 21 and 23

A new antibiotic is actively secreted by the kidney. Its V_D is 25 L in the normal adult and its clearance is 750 mL/min. What is its elimination rate constant in reciprocal hours?

Slides 21 and 23

A new antibiotic is actively secreted by the kidney; V_D is 25 L in the normal adult. The clearance of this drug is 750 mL/min. What is the usual t½ for this drug?

Slides 21 and 23

The same antibiotic has a V_D of 25 L. What would be the new t½ for this drug in an adult with partial renal failure whose clearance of the antibiotic was 150 mL/min?

Slide 17 A patient is 165 cm tall. What is this height in inches?

Slides 16 and 17

A female patient is 165 cm tall. What is her ideal body weight?

Slide 17 Estimate, using the ideal body weight, the creatinine clearance of a 45-year-old female who weighs 61 kg, has a serum creatinine of 1.1 mg/dL, and is 165 cm tall.